

Convergence of exact Nash equilibria to mean field equilibria in ergodic Markovian games

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Website #9 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

For each n , we are given a symmetric continuous-time n -player stochastic game on the finite state space $[d] = \{1, \dots, d\}$, with admissible jump rates bounded away from 0 and ∞ . The question asks for conditions under which *exact* Markovian Nash equilibria of the finite- n ergodic game converge, as $n \rightarrow \infty$, to the corresponding ergodic mean field game (MFG) equilibrium, in the sense of convergence of equilibrium empirical measures (or their stationary laws) and convergence of equilibrium average costs to the MFG ergodic value.

I will prove a rigorous stationary-version theorem: for a sequence of *symmetric stationary closed-loop exact Nash equilibria* whose feedbacks have a stable mean-field limit, the invariant law of the equilibrium empirical measure converges to a stationary MFG equilibrium $\bar{\mu}$, and the common equilibrium values converge to the stationary MFG value ρ .

Alternative readings considered

The statement admits at least two readings.

- (R1) The most ambitious reading allows arbitrary exact Markovian equilibria, possibly time-dependent and not permutation-equivariant.
- (R2) The stationary ergodic reading restricts attention to symmetric stationary closed-loop equilibria of empirical-measure type, which is the natural setting for convergence of invariant laws and stationary MFG equilibria.

I solve (R2). I do *not* solve the fully time-dependent closed-loop problem of (R1); that is recorded in the gap section.

Related literature found

- **PARTIAL**: Cecchin–Pelino (2019) proves convergence of finite-state symmetric N -player *finite-horizon* feedback Nash equilibria, values, and trajectories to the finite-state MFG by exploiting regularity of the master equation. This is the closest finite-state closed-loop neighbor, but it is not an ergodic average-cost result.
- **PARTIAL**: Bayraktar–Cohen (2018) studies a finite-state many-player game on a finite horizon and shows that the n -player value functions are approximated by the master equation; it is a useful nearby model, but again not the ergodic exact-equilibrium converse asked here.
- **BACKGROUND/TOOL**: Cohen–Zell (2022) analyzes the finite-state ergodic master equation and, under Lasry–Lions monotonicity, links the stationary ergodic MFG equilibrium to the ergodic master equation. This provides the canonical candidate limit feedback in the ergodic regime.

- **PARTIAL:** Cohen–Zell (2024) proves that an ergodic finite-state MFG solution yields a $C/\sqrt{n}$ -Nash equilibrium in the n -player game, and likewise for the master-equation-derived profile. This is the direction “MFG $\Rightarrow$ asymptotic Nash,” not the converse “exact Nash $\Rightarrow$ MFG limit.”
- **PARTIAL:** Doncel–Gast–Gaujal (2019) treats discrete-state mean field games in continuous/discrete time and finite/infinite horizon, proves convergence for a restricted strategy class, and also exhibits equilibria outside that class that do *not* converge to mean field equilibria. This shows extra structural assumptions are genuinely necessary.
- **PARTIAL:** Cecchin–Dai Pra–Fischer–Pelino (2019) studies a finite-horizon two-state model without uniqueness and shows a selection phenomenon in the limit. This strongly suggests that uniqueness/stability assumptions cannot simply be ignored in convergence problems.

The web search therefore suggested the following strategy: do *not* try to prove a sweeping converse theorem for arbitrary exact ergodic closed-loop equilibria; instead isolate a robust stationary setting where one can prove convergence directly from (i) exact Nash optimality, (ii) a stable limiting feedback, and (iii) the fact that a single deviating player does not move the empirical measure of the $(n - 1)$ *other* players except through an $O(1/n)$ perturbation.

Proof sketch

Write

$$N^n := \mu^{n,1} = \frac{1}{n-1} \sum_{j=2}^n \delta_{X^j}$$

for the empirical measure of the players other than player 1. The key observation is that, under either the equilibrium profile or a one-player deviation, the process N^n changes *only* when players $2, \dots, n$ jump. Player 1’s jumps do not move N^n at all. They enter only through the arguments of the feedbacks used by players $2, \dots, n$, and this influence is of size $1/(n - 1)$.

Assume the equilibrium feedbacks converge to a limiting feedback ϕ , and assume the McKean–Vlasov drift $b(\eta) = \eta Q_\phi(\eta)$ admits a strict C^2 Lyapunov function V with a unique zero at $\bar{\mu}$. Then a generator expansion for $V(N^n)$ shows that, under the invariant law of the equilibrium profile *and also under the invariant law of any one-player deviation*, the empirical measure N^n satisfies

$$\mathbb{E}|N^n - \bar{\mu}|^2 \leq C \left(\varepsilon_n + \frac{1}{n} \right), \quad \varepsilon_n := \sup |\phi^{(n)} - \phi|.$$

Thus the environment seen by a tagged player concentrates around $\bar{\mu}$, both at equilibrium and after a single deviation.

Next, because the finite- n equilibrium is exact Nash, player 1 cannot improve by switching to any stationary control β depending only on her own state. Under such a deviation, player 1’s marginal chain is autonomous with generator Q_β , while the environment N^n still concentrates at $\bar{\mu}$. Hence the deviating cost converges to the frozen-environment ergodic cost

$$\mathcal{J}_{\bar{\mu}}(\beta) = \sum_x \pi_x^\beta (f(x, \beta(x)) + F(x, \bar{\mu})),$$

where π^β is the invariant law of Q_β .

Passing to the limit in the exact Nash inequalities yields

$$\mathcal{J}_{\bar{\mu}}(\phi(\cdot, \bar{\mu})) \leq \mathcal{J}_{\bar{\mu}}(\beta) \quad \forall \beta.$$

A second generator identity shows $\bar{\mu}Q_{\phi(\cdot, \bar{\mu})} = 0$. Therefore $(\bar{\mu}, \phi(\cdot, \bar{\mu}))$ is a stationary MFG equilibrium. The equilibrium values ρ_n then converge to

$$\varrho = \mathcal{J}_{\bar{\mu}}(\phi(\cdot, \bar{\mu})) = \inf_{\beta} \mathcal{J}_{\bar{\mu}}(\beta).$$

Main result

For a stationary own-state feedback β , let Q_{β} be the $d \times d$ rate matrix with off-diagonal entries $\beta_y(x) \in [\underline{a}, \bar{a}]$ for $y \neq x$, and diagonal $\beta_x(x) = -\sum_{y \neq x} \beta_y(x)$. Let π^{β} denote its invariant law. For $\eta \in \mathcal{P}([d])$, define

$$\mathcal{J}_{\eta}(\beta) := \sum_{x=1}^d \pi_x^{\beta} \left(f(x, \beta(x)) + F(x, \eta) \right).$$

I call (η, β) a *stationary MFG equilibrium* if

$$\eta Q_{\beta} = 0 \quad \text{and} \quad \mathcal{J}_{\eta}(\beta) \leq \mathcal{J}_{\eta}(\beta') \quad \text{for all stationary own-state feedbacks } \beta'.$$

Theorem 1 (Stable-limit convergence of exact ergodic Nash equilibria). *Assume:*

(H1) *For each n , $\alpha^{(n)}$ is a symmetric stationary exact Nash equilibrium of empirical-measure form:*

$$\alpha_y^{(n),i}(x) = \phi_y^{(n)}(x_i, \mu^{n,i}(x)), \quad y \neq x_i,$$

for some map $\phi^{(n)} : [d] \times \mathcal{P}([d]) \rightarrow [\underline{a}, \bar{a}]^{d-1}$.

(H2) *$f(x, \cdot)$ is bounded and Lipschitz on the compact action set, uniformly in x , and $F(x, \cdot)$ is bounded and Lipschitz on $\mathcal{P}([d])$, uniformly in x .*

(H3) *There exists $L < \infty$ such that each $\phi^{(n)}$ is L -Lipschitz in the measure variable, uniformly in x . Moreover $\phi^{(n)} \rightarrow \phi$ uniformly on $[d] \times \mathcal{P}([d])$; write*

$$\varepsilon_n := \sup_{x, \eta, y \neq x} |\phi_y^{(n)}(x, \eta) - \phi_y(x, \eta)| \xrightarrow{n \rightarrow \infty} 0.$$

(H4) *Let $Q_{\phi}(\eta)$ be the rate matrix with off-diagonal entries $\phi_y(x, \eta)$, and define the McKean–Vlasov drift*

$$b(\eta) := \eta Q_{\phi}(\eta) \in \mathbb{R}^d.$$

There exist $\bar{\mu} \in \mathcal{P}([d])$, $c_0 > 0$, and a function $V \in C^2$ on a neighborhood of $\mathcal{P}([d])$ such that

$$\nabla V(\eta) \cdot b(\eta) \leq -c_0 |\eta - \bar{\mu}|^2 \quad \text{for all } \eta \in \mathcal{P}([d]).$$

Let $\rho_n := J_0^{n,1}(\alpha^{(n)})$, which is the common equilibrium value by symmetry. Then:

(1) *If Π^n is the invariant law of the n -player chain under $\alpha^{(n)}$, and*

$$M^n := \frac{1}{n} \sum_{i=1}^n \delta_{X^i}, \quad N^n := \mu^{n,1} = \frac{1}{n-1} \sum_{j=2}^n \delta_{X^j},$$

then

$$\mathbb{E}_{\Pi^n} |N^n - \bar{\mu}|^2 \leq C \left(\varepsilon_n + \frac{1}{n} \right), \quad \mathbb{E}_{\Pi^n} |M^n - \bar{\mu}|^2 \leq C \left(\varepsilon_n + \frac{1}{n} \right),$$

for a constant C independent of n . In particular $M^n \rightarrow \bar{\mu}$ in probability under Π^n . Hence, if each chain is started from Π^n , the stationary empirical-measure process converges in finite-dimensional distributions to the constant path $t \mapsto \bar{\mu}$.

(2) The pair $(\bar{\mu}, \phi(\cdot, \bar{\mu}))$ is a stationary MFG equilibrium:

$$\bar{\mu}Q_{\phi(\cdot, \bar{\mu})} = 0, \quad \mathcal{J}_{\bar{\mu}}(\phi(\cdot, \bar{\mu})) = \inf_{\beta} \mathcal{J}_{\bar{\mu}}(\beta) =: \varrho.$$

(3) The equilibrium values converge:

$$\rho_n \longrightarrow \varrho.$$

More precisely,

$$|\rho_n - \varrho| \leq C \left(\varepsilon_n + \sqrt{\varepsilon_n + \frac{1}{n}} \right).$$

(4) If, in addition, the ergodic master equation has a smooth solution U_0 with minimizing selector γ^* , and if

$$\phi(x, \eta) = \gamma^*(x, \Delta_x U_0(\cdot, \eta)),$$

then $(\bar{\mu}, \phi(\cdot, \bar{\mu}))$ is exactly the stationary MFG equilibrium generated by the master equation. If that stationary MFG equilibrium is unique, then the finite- n exact equilibria converge to that unique mean field equilibrium.

Proof

Fix n . Because every off-diagonal rate is in $[\underline{a}, \bar{a}]$ with $\underline{a} > 0$, from any configuration $x \in [d]^n$ one can reach any other configuration $x' \in [d]^n$ by changing coordinates one at a time, and every required move has positive rate. Thus every stationary profile considered below induces an irreducible continuous-time Markov chain on the finite state space $[d]^n$. Consequently it has a unique invariant law, and the long-run average cost equals the expectation of the running cost under that invariant law. I use this repeatedly without further comment.

Let $e_1, \dots, e_d$ be the standard basis of $\mathbb{R}^d$.

Lemma 1 (Concentration of the empirical measure of the $(n-1)$ other players). *Let β be any stationary Markovian control for player 1 (possibly depending on the whole configuration). Let $\tilde{\Pi}^{n, \beta}$ be the invariant law of the profile $(\alpha^{(n), -1}; \beta)$, and let*

$$N^n = \frac{1}{n-1} \sum_{j=2}^n \delta_{X^j}.$$

Then

$$\mathbb{E}_{\tilde{\Pi}^{n, \beta}} |N^n - \bar{\mu}|^2 \leq C \left(\varepsilon_n + \frac{1}{n} \right),$$

with a constant C independent of n and of β .

Proof. Consider the function $G(x) := V(N^n(x))$ on $[d]^n$. Under the profile $(\alpha^{(n), -1}; \beta)$, only jumps of players $2, \dots, n$ change N^n . Hence, if $\mathcal{L}^{n, \beta}$ denotes the generator,

$$(\mathcal{L}^{n, \beta} G)(x) = \sum_{j=2}^n \sum_{y \neq x_j} \phi_y^{(n)}(x_j, \mu^{n, j}(x)) \left[V \left(N^n(x) + \frac{e_y - e_{x_j}}{n-1} \right) - V(N^n(x)) \right].$$

Now

$$\mu^{n, j}(x) = \frac{1}{n-1} \sum_{k \neq j} \delta_{x_k} = N^n(x) + \frac{e_{x_1} - e_{x_j}}{n-1},$$

so

$$|\mu^{n,j}(x) - N^n(x)| \leq \frac{|e_{x_1} - e_{x_j}|}{n-1} \leq \frac{2}{n-1}.$$

By the second-order Taylor formula, for any $\eta \in \mathcal{P}([d])$ and any h with $\eta + h \in \mathcal{P}([d])$,

$$V(\eta + h) - V(\eta) = \nabla V(\eta) \cdot h + R(\eta, h), \quad |R(\eta, h)| \leq C_V |h|^2,$$

where $C_V := \frac{1}{2} \sup_{\zeta} \|D^2 V(\zeta)\|$. Applying this with

$$h = \frac{e_y - e_{x_j}}{n-1}$$

gives

$$(\mathcal{L}^{n,\beta} G)(x) = \nabla V(N^n(x)) \cdot B^n(x) + \mathcal{R}^n(x),$$

where

$$B^n(x) := \frac{1}{n-1} \sum_{j=2}^n \sum_{y \neq x_j} \phi_y^{(n)}(x_j, \mu^{n,j}(x)) (e_y - e_{x_j}),$$

and, since each player jumps at total rate at most $(d-1)\bar{a}$,

$$|\mathcal{R}^n(x)| \leq \frac{C}{n-1}.$$

Next compare $B^n(x)$ with the limiting drift

$$b(\eta) = \sum_{z=1}^d \eta_z \sum_{y \neq z} \phi_y(z, \eta) (e_y - e_z).$$

Because

$$\frac{1}{n-1} \sum_{j=2}^n \sum_{y \neq x_j} \phi_y(x_j, N^n(x)) (e_y - e_{x_j}) = b(N^n(x)),$$

we have

$$\begin{aligned} |B^n(x) - b(N^n(x))| &\leq \frac{1}{n-1} \sum_{j=2}^n \sum_{y \neq x_j} |e_y - e_{x_j}| \left| \phi_y^{(n)}(x_j, \mu^{n,j}(x)) - \phi_y(x_j, N^n(x)) \right| \\ &\leq C \left(\varepsilon_n + \frac{1}{n} \right), \end{aligned}$$

because $|e_y - e_{x_j}| \leq 2$, the family $\phi^{(n)}$ is uniformly Lipschitz in the measure argument, and

$$|\mu^{n,j}(x) - N^n(x)| \leq \frac{2}{n-1}.$$

Therefore

$$(\mathcal{L}^{n,\beta} G)(x) \leq \nabla V(N^n(x)) \cdot b(N^n(x)) + C \left(\varepsilon_n + \frac{1}{n} \right).$$

By the Lyapunov assumption (H4),

$$(\mathcal{L}^{n,\beta} G)(x) \leq -c_0 |N^n(x) - \bar{\mu}|^2 + C \left(\varepsilon_n + \frac{1}{n} \right).$$

Finally, under the invariant law $\tilde{\Pi}^{n,\beta}$,

$$0 = \mathbb{E}_{\tilde{\Pi}^{n,\beta}}[(\mathcal{L}^{n,\beta}G)(X)],$$

hence

$$c_0 \mathbb{E}_{\tilde{\Pi}^{n,\beta}} |N^n - \bar{\mu}|^2 \leq C \left(\varepsilon_n + \frac{1}{n} \right),$$

which is the claim. $\square$

Lemma 2 (Identification of the stationary limit at equilibrium). *Let Π^n be the invariant law of the equilibrium profile $\alpha^{(n)}$. Then:*

(a) $\mathbb{E}_{\Pi^n} N^n = \text{Law}_{\Pi^n}(X^1)$, hence $\text{Law}_{\Pi^n}(X^1) \rightarrow \bar{\mu}$.

(b) $\bar{\mu} Q_{\phi(\cdot, \bar{\mu})} = 0$.

(c) If $\bar{\rho} := \mathcal{J}_{\bar{\mu}}(\phi(\cdot, \bar{\mu}))$, then

$$|\rho_n - \bar{\rho}| \leq C \left(\varepsilon_n + \sqrt{\varepsilon_n + \frac{1}{n}} \right).$$

Proof. Because the equilibrium profile is symmetric and the invariant law is unique, Π^n is exchangeable. Therefore, for each $z \in [d]$,

$$\mathbb{E}_{\Pi^n}[N_z^n] = \frac{1}{n-1} \sum_{j=2}^n \Pi^n(X^j = z) = \Pi^n(X^1 = z).$$

Thus $\mathbb{E}_{\Pi^n} N^n = \text{Law}_{\Pi^n}(X^1)$. By Jensen and the previous lemma,

$$|\text{Law}_{\Pi^n}(X^1) - \bar{\mu}|^2 = |\mathbb{E}_{\Pi^n} N^n - \bar{\mu}|^2 \leq \mathbb{E}_{\Pi^n} |N^n - \bar{\mu}|^2 \leq C \left(\varepsilon_n + \frac{1}{n} \right).$$

This proves (a).

For (b), let $\psi : [d] \rightarrow \mathbb{R}$. Under the equilibrium profile,

$$0 = \mathbb{E}_{\Pi^n} \left[\sum_{y \neq X^1} \phi_y^{(n)}(X^1, N^n) (\psi(y) - \psi(X^1)) \right]$$

because Π^n is invariant and the displayed integrand is the generator applied to the observable $x \mapsto \psi(x_1)$.

Set

$$g_n(x, \eta) := \sum_{y \neq x} \phi_y^{(n)}(x, \eta) (\psi(y) - \psi(x)), \quad g(x, \eta) := \sum_{y \neq x} \phi_y(x, \eta) (\psi(y) - \psi(x)).$$

Since $\phi^{(n)} \rightarrow \phi$ uniformly and ϕ is continuous, for each x ,

$$|g_n(x, \eta) - g(x, \bar{\mu})| \leq C(\varepsilon_n + |\eta - \bar{\mu}|).$$

Hence

$$\left| \mathbb{E}_{\Pi^n}[g_n(X^1, N^n)] - \sum_{x=1}^d \Pi^n(X^1 = x) g(x, \bar{\mu}) \right| \leq C \left(\varepsilon_n + \mathbb{E}_{\Pi^n} |N^n - \bar{\mu}| \right) \rightarrow 0.$$

Using part (a), $\Pi^n(X^1 = x) \rightarrow \bar{\mu}_x$. Passing to the limit in

$$\mathbb{E}_{\Pi^n}[g_n(X^1, N^n)] = 0$$

gives

$$\sum_{x=1}^d \bar{\mu}_x \sum_{y \neq x} \phi_y(x, \bar{\mu}) (\psi(y) - \psi(x)) = 0.$$

Since this holds for every ψ , it is exactly the statement $\bar{\mu} Q_{\phi(\cdot, \bar{\mu})} = 0$. This proves (b).

For (c), by invariance of Π^n ,

$$\rho_n = \mathbb{E}_{\Pi^n} \left[f(X^1, \phi^{(n)}(X^1, N^n)) + F(X^1, N^n) \right].$$

Since f and F are Lipschitz and bounded,

$$\left| f(X^1, \phi^{(n)}(X^1, N^n)) + F(X^1, N^n) - (f(X^1, \phi(X^1, \bar{\mu})) + F(X^1, \bar{\mu})) \right| \leq C(\varepsilon_n + |N^n - \bar{\mu}|).$$

Taking expectation and then replacing the law of X^1 by $\bar{\mu}$ using part (a),

$$|\rho_n - \bar{\rho}| \leq C \left(\varepsilon_n + \mathbb{E}_{\Pi^n} |N^n - \bar{\mu}| + |\text{Law}_{\Pi^n}(X^1) - \bar{\mu}| \right).$$

By the previous lemma and part (a), the right-hand side is bounded by

$$C \left(\varepsilon_n + \sqrt{\varepsilon_n + \frac{1}{n}} \right).$$

Finally, part (b) shows that $\bar{\mu}$ is the invariant law of $Q_{\phi(\cdot, \bar{\mu})}$, so indeed

$$\bar{\rho} = \mathcal{J}_{\bar{\mu}}(\phi(\cdot, \bar{\mu})).$$

□

Lemma 3 (Deviation costs converge to the frozen-environment costs). *Let β be any stationary own-state feedback. Let $\tilde{\Pi}^{n, \beta}$ be the invariant law of $(\alpha^{(n), -1}; \beta)$, and let*

$$J_n^\beta := J_0^{n, 1}(\alpha^{(n), -1}; \beta).$$

Then

$$|J_n^\beta - \mathcal{J}_{\bar{\mu}}(\beta)| \leq C \sqrt{\varepsilon_n + \frac{1}{n}},$$

and in particular $J_n^\beta \rightarrow \mathcal{J}_{\bar{\mu}}(\beta)$.

Proof. Under the deviated profile $(\alpha^{(n), -1}; \beta)$, player 1's jump rates depend only on her own state. Hence, if $m_n^\beta := \text{Law}_{\tilde{\Pi}^{n, \beta}}(X^1)$, then for every $\psi : [d] \rightarrow \mathbb{R}$,

$$0 = \mathbb{E}_{\tilde{\Pi}^{n, \beta}} \left[\sum_{y \neq X^1} \beta_y(X^1) (\psi(y) - \psi(X^1)) \right] = \sum_{x=1}^d m_n^\beta(x) \sum_{y \neq x} \beta_y(x) (\psi(y) - \psi(x)).$$

Thus $m_n^\beta Q_\beta = 0$. Since all off-diagonal rates of Q_β lie in $[a, \bar{a}]$, Q_β is irreducible on $[d]$, so its invariant law is unique. Therefore

$$m_n^\beta = \pi^\beta.$$

Again by invariance of $\tilde{\Pi}^{n,\beta}$,

$$J_n^\beta = \sum_{x=1}^d \pi_x^\beta f(x, \beta(x)) + \mathbb{E}_{\tilde{\Pi}^{n,\beta}}[F(X^1, N^n)].$$

By the first lemma,

$$\mathbb{E}_{\tilde{\Pi}^{n,\beta}}|N^n - \bar{\mu}|^2 \leq C\left(\varepsilon_n + \frac{1}{n}\right),$$

hence

$$\mathbb{E}_{\tilde{\Pi}^{n,\beta}}|N^n - \bar{\mu}| \leq C\sqrt{\varepsilon_n + \frac{1}{n}}.$$

Using the Lipschitz continuity of F ,

$$\left| \mathbb{E}_{\tilde{\Pi}^{n,\beta}}[F(X^1, N^n)] - \sum_{x=1}^d \pi_x^\beta F(x, \bar{\mu}) \right| \leq C\sqrt{\varepsilon_n + \frac{1}{n}}.$$

Combining the last two displays yields

$$|J_n^\beta - \mathcal{J}_\mu(\beta)| \leq C\sqrt{\varepsilon_n + \frac{1}{n}}.$$

□

Proof of the theorem. Item (1) for N^n is the first lemma with $\beta = \alpha^{(n),1}$. Since

$$M^n = \frac{1}{n}\delta_{X^1} + \frac{n-1}{n}N^n,$$

we have

$$|M^n - \bar{\mu}| \leq |N^n - \bar{\mu}| + \frac{2}{n},$$

so the same estimate holds for M^n . If the chain is started from Π^n , then it is stationary, hence M_t^n has the same law as M_0^n for every fixed t . Therefore each fixed-time marginal converges in probability to $\bar{\mu}$, and the finite-dimensional convergence to the constant path follows from a union bound.

Now fix any stationary own-state feedback β . Since $\alpha^{(n)}$ is an exact Nash equilibrium,

$$\rho_n \leq J_n^\beta.$$

Pass to the limit using the previous two lemmas:

$$\mathcal{J}_\mu(\phi(\cdot, \bar{\mu})) = \lim_{n \rightarrow \infty} \rho_n \leq \lim_{n \rightarrow \infty} J_n^\beta = \mathcal{J}_\mu(\beta).$$

Because β was arbitrary,

$$\mathcal{J}_\mu(\phi(\cdot, \bar{\mu})) \leq \inf_{\beta} \mathcal{J}_\mu(\beta).$$

The reverse inequality is automatic because $\phi(\cdot, \bar{\mu})$ itself is an admissible stationary own-state feedback. Hence

$$\mathcal{J}_\mu(\phi(\cdot, \bar{\mu})) = \inf_{\beta} \mathcal{J}_\mu(\beta) =: \varrho.$$

Together with Lemma 2(b), this proves that $(\bar{\mu}, \phi(\cdot, \bar{\mu}))$ is a stationary MFG equilibrium. This is item (2).

Item (3) now follows from Lemma 2(c), because $\bar{\rho} = \mathcal{J}_{\bar{\mu}}(\phi(\cdot, \bar{\mu})) = \varrho$:

$$|\rho_n - \varrho| = |\rho_n - \bar{\rho}| \leq C\left(\varepsilon_n + \sqrt{\varepsilon_n + \frac{1}{n}}\right).$$

For item (4), assume additionally that

$$\phi(x, \eta) = \gamma^*(x, \Delta_x U_0(\cdot, \eta))$$

for a smooth ergodic master-equation solution U_0 . Then the optimality and consistency already proved show that the stationary pair $(\bar{\mu}, \phi(\cdot, \bar{\mu}))$ is precisely the stationary MFG equilibrium generated by the master-equation feedback. If that equilibrium is unique, there is no subsequential ambiguity, so the finite- n exact equilibria converge to that unique MFG equilibrium. $\square$

Gap to the full statement

What remains open in my argument is precise and concrete.

- I do *not* handle arbitrary time-dependent exact Markovian equilibria. My proof is stationary and invariant-law based.
- I assume the exact equilibria are symmetric and of empirical-measure feedback form

$$\alpha^{(n),i}(x) = \phi^{(n)}(x_i, \mu^{n,i}(x)).$$

I do not treat general closed-loop equilibria depending on the whole configuration in a non-exchangeable way.

- I assume a limiting feedback ϕ exists and has a strict Lyapunov-stable stationary point $\bar{\mu}$. The proof does not show that these properties follow automatically from mere existence of exact equilibria.
- I prove convergence to a *stationary* MFG equilibrium. I do not prove convergence to a non-stationary MFG flow $\mu(t)$.

These restrictions are not cosmetic. The literature contains discrete-state examples where convergence fails outside suitably restricted strategy classes, and finite-horizon examples where nonuniqueness leads to selection phenomena rather than unconditional convergence.

Self red-team

Counterexample attempts. I tried to break the theorem by removing each major hypothesis.

- If the Lyapunov condition is dropped, it is easy to imagine two distinct stable fixed points of the limiting drift b . Then one can have subsequences of equilibria converging to different limits, so full-sequence convergence should fail.
- If the empirical-feedback structure is dropped, a closed-loop equilibrium may encode much richer dependence on the full configuration; the generator estimate for N^n then no longer closes.
- If the lower bound $\underline{a} > 0$ is dropped, irreducibility can fail and the ergodic cost can depend on the initial condition. Then the whole invariant-law method collapses.

Three plausible referee objections.

- *Objection 1:* “A deviating player could affect the other players over long times, so why does N^n still stay near $\bar{\mu}$?”

Answer: I never compare the deviated system to the equilibrium system. I estimate N^n directly *under the deviated invariant law*. Since player 1’s jumps do not move N^n , and enter the other players’ feedbacks only through an $O(1/n)$ perturbation of empirical arguments, the same Lyapunov calculation still works.

- *Objection 2:* “Why is $\bar{\rho}$ the ergodic cost of the limit feedback in frozen environment $\bar{\mu}$?”

Answer: This requires the consistency identity $\bar{\mu}Q_{\phi(\cdot, \bar{\mu})} = 0$, which I proved separately in Lemma 2(b) by applying the generator to test functions of the tagged player’s state.

- *Objection 3:* “You use only own-state deviations β , whereas exact Nash allows all Markovian deviations.”

Answer: Exact Nash dominates every Markovian deviation, hence in particular this subclass. That subclass is enough to identify the stationary frozen-environment ergodic control problem and therefore enough to prove that the limit pair is a stationary MFG equilibrium.

Boundary cases checked.

- If $d = 1$, then $M^n \equiv \bar{\mu} = \delta_1$, and the theorem is trivially true.
- If F does not depend on the measure, the result reduces to convergence toward the invariant law of the limiting controlled Markov chain.
- The proof uses every structural hypothesis: $\underline{a} > 0$ for irreducibility, Lipschitz control feedbacks for the $O(1/n)$ perturbation estimate, and the Lyapunov inequality for concentration.

Comparison with known failure modes. The theorem does not pretend to hold without stability/structure assumptions. That would be implausible in view of discrete-state nonconvergence examples outside well-behaved strategy classes and finite-horizon selection phenomena under nonuniqueness.

Ideas tried

- **Approach A: finite-player Nash system versus projected master equation.**

Idea: write the stationary Nash system for the exact n -player equilibrium, project the master equation onto $[d]^n$, and prove the exact Nash system is a perturbation of that projected equation.

Obstruction: in the ergodic stationary setting one needs a uniqueness/comparison mechanism for the stationary n -player Nash system, and that mechanism is not available from the problem statement alone. This route would have required importing a large amount of external theory.

Next step if pursued: impose a monotonicity/contractivity condition strong enough to make the stationary Nash system uniquely solvable and quantitatively stable.

- **Approach B: exchangeability plus de Finetti.**

Idea: use exchangeability of symmetric equilibria to extract subsequential limits of the invariant laws and identify the de Finetti measure as a mixture of stationary MFG equilibria.

Obstruction: this only gives a *mixture* statement. It does not force concentration at a deterministic $\bar{\mu}$ and does not by itself give value convergence. A stability input is still needed.

Next step if pursued: combine de Finetti with a strict Lyapunov or strict monotonicity condition to collapse the directing measure to a Dirac mass.

- **Approach C: direct coupling between the equilibrium system and a one-player-deviation system.**

Idea: couple the two n -player chains and estimate the Hamming distance between the two empirical measures.

Obstruction: a naive Grönwall bound produces a creation term proportional to the current discrepancy. To close it uniformly in time one needs a strong contractivity estimate, which is not automatic.

Next step if pursued: impose a Dobrushin-type contraction condition on the feedback map.

- **Approach D (successful): Lyapunov analysis of the $(n-1)$ -player environment seen by a tagged player.**

Idea: study $N^n = \mu^{n,1}$ directly. Because player 1's jumps do not move N^n , the environment seen by the tagged player remains governed by the same stable mean-field drift up to an $O(1/n)$ perturbation, even after a deviation.

Why it worked: it uses exact Nash only at the very end, where it belongs, and keeps the probabilistic part of the proof entirely on the environment process.

Invoked results

No external theorem is used as a black box in the proof. The web sources were used only to map the nearby literature and to check what is known and what is not known. Every step of the proof itself is derived in the manuscript.

Self-confidence

3. I checked every nontrivial step and the argument is internally consistent, but the result is a deliberately structured partial theorem rather than the full general convergence statement.

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